Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: Claim 3 and the missing lift. The finite-root example satisfies the stated conclusions of HK, Proposition 10.2, Claim 3, after correcting the swapped names of its factor maps. The missing conclusion is stronger: the Γ factor must have an RV lift over the original ground to be a morphism of the restricted Γ category used in the tensor comparison.

Write h for the RES factor and k for the Γ factor. Take $\mathbb{S} = \mathbb{C}((t))$, $\mathrm{val}(t) = 1$, and

$$\delta = \tfrac{1}{5}, \qquad R = \{r \in \mathrm{RV} : r^5 = \mathrm{rv}(t)\}, \qquad I = (0, \infty), \quad J = (\delta, \infty).$$

Then R is a five-element RES[0] object and a complete algebraic type over $\mathbb{S}$: the roots are permuted transitively over this ground. Consider the original-ground definable bijection

$$F : R \times I^\sharp \longrightarrow R \times J^\sharp, \qquad F(r, u) = (r, ru),$$

with inverse $(r, w) \mapsto (r, r^{-1}w)$. Let P be the complete Γ type at positive infinity over $\mathbb{S}$, and put $Q = R$, $Q' = R$, $P' = P + \delta$. All of Claim 3's factor conclusions hold:

$$F(Q \times P^\sharp) = Q' \times (P')^\sharp, \qquad h(r) = r, \quad k(\gamma) = \gamma + \delta,$$

$$F(\{r\} \times \gamma^\sharp) = \{r\} \times (\gamma + \delta)^\sharp.$$

The definable neighbourhoods can initially be R, I, R, J, with h and k original-ground definable bijections. These factor maps are forced by the first coordinate and the output value of F.

The base of the lift. For each fixed r, the map $L_r(u) = ru$ lifts k, but is defined over $\mathbb{S}(r)$. With r retained as an input, F is a definable family of such lifts. Independence of the formula for k from r does not produce a lift defined over $\mathbb{S}$. In particular, the implication

$$[\text{for every } r \in R,\ k \text{ has a lift over } \mathbb{S}(r)] \implies [k \text{ has a lift over } \mathbb{S}]$$

is false here.

Shrinking does not fix it. Every $\mathbb{S}$-definable Γ neighbourhood U of the type P contains a final interval (c, ∞) with $c \in \mathbb{Q}$. Choose an integer $N > \max(c, 0)$. Then $u_N = \mathrm{rv}(t)^N$ is an $\mathbb{S}$-definable point of $U^\sharp$. Any $\mathbb{S}$-definable function L contracting to k would satisfy

$$\mathrm{vrv}(L(u_N)) = N + \tfrac{1}{5}.$$

But the henselian ground has $\mathrm{dcl}(\mathbb{S}) \cap \mathrm{RV} = \mathrm{RV}(\mathbb{S})$, whose value group is $\mathbb{Z}$. Thus no such L exists, even without requiring bijectivity. Nor can a residual neighbourhood select one root: R itself is a complete algebraic type over $\mathbb{S}$.

Exact location of the gap. Clause (3) of Claim 3 gives a lift after the RES parameter is fixed. It does not establish the same-ground lift needed to use k as a Γ morphism. The tensor equality asserted after Claim 4 therefore needs an additional justification in the restricted category or in the balanced tensor; it does not follow merely from the stated definable factor bijections. The earlier T001 detector addresses the balancing relations and distinguishes the global tensors for R, I, J. Compactness can give neighbourhoods and a uniform family retaining r; it cannot remove this parameter. This diagnoses an insufficient Claim 3 for the induction, rather than a failure of its literal fibre equations in this example. If Claim 3 intended a Γ morphism rather than merely a definable Γ bijection, that stronger reading is false here. No change to the working manuscript is proposed here. The concrete model and complete tensor detector have not yet been constructed in Lean.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of RV[$*$] then it may also denote the object $(\boldsymbol{U}, 0)$ of μRV[$*$] with the constant volume form 0.

Notation 3.20. Let $[1] \in \mathbf{K}_+\,\mathrm{RES}[1]$ be the class of the singleton $\{1\}$. The class of the singleton $\{1\}$ in $\mathbf{K}_+\,\mathrm{RES}[0]$ is the multiplicative identity of $\mathbf{K}_+\,\mathrm{RES}[*]$ and hence is simply denoted by 1. We have $[\mathrm{RV}^{\circ\circ}_\infty] = [\mathrm{RV}^{\circ\circ}] + 1$ in $\mathbf{K}_+\,\mathrm{RV}[\leq 1]$.